

PROJECT DOCUMENTATION OF **REVIEW - 1**

AI-POWERED BANKING CUSTOMER COMPLAINT MANAGEMENT SYSTEM USING BERT

Team Number: 14

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Problem Statement

- Banks receive a large volume of customer complaints every day across different products and services, making manual complaint classification and routing a time-consuming and inefficient process.
- Customer complaints are often written in natural language and may contain multiple issues, incomplete information, or different expressions for the same problem, making it difficult to accurately identify the complaint category using simple keyword matching.
- Traditional machine learning approaches such as Logistic Regression, Naïve Bayes, and Support Vector Machines generally depend on features such as TF-IDF and Bag-of-Words, which may not capture the deeper semantic and contextual meaning of complaint narratives.
- Incorrect classification of complaints can result in routing them to the wrong department, increasing processing time and potentially delaying customer resolution.
- Manual prioritization also makes it difficult to consistently identify urgent or high-impact complaints that require immediate attention.
- Therefore, there is a need for an intelligent complaint management system that can automatically analyze complaint descriptions, understand their contextual meaning, classify them into appropriate categories, determine their priority, and route them to the relevant department.

Existing System

Existing Approach	Limitation
Manual Complaint Processing	Time-consuming and inconsistent
Keyword-Based Routing	Cannot understand semantic context
Logistic Regression	Relies on TF-IDF/Bag-of-Words features
Naïve Bayes	Limited contextual understanding
Support Vector Machine (SVM)	Requires handcrafted feature engineering

Literature Survey

Paper / Year	Method Used	Dataset	What They Did	Limitation
Bastani et al. (2019)	LDA	CFPB	Found common topics in consumer complaints	Focuses on topics, not detailed complaint classification or management
Sangaraju et al. (2022)	BERT Embeddings + BERTopic	CFPB	Used BERT-based representations to find complaint topics	Mainly focuses on topic analysis, not complete complaint handling
Vaishnav et al. (2024)	Machine Learning + LDA	CFPB	Used complaint data for analysis and prediction	Does not provide a complete complaint routing system
Marshan et al. (2023)	NLP + Emotion Detection	Customer complaint data	Analyzed customer complaints and emotions	Focuses on emotion/priority, not the complete complaint lifecycle
Mokase et al. (2025)	LSTM vs BERT	Complaint dataset	Compared LSTM and BERT for complaint classification	Mainly focuses on classification rather than complaint management
Casanueva et al. (2020)	Intent Classification	BANKING77	Classified banking customer queries into 77 intents	Focuses on intent classification, not detailed complaint management
Jarrar et al. (2023)	AraBERT	ArBanking77	Applied BERT to Arabic banking intent classification	Focuses on Arabic intent detection
Gao et al. (2023)	NLP + Anomaly Detection	CFPB	Used complaint text to detect unusual complaint patterns	Focuses on anomaly detection rather than complaint routing
Towards Automated Customer	NLP / Neural Models	Customer Support on Twitter	Studied automated responses for customer-support conversations	Focuses on support conversations rather than banking complaint management

Support (2018)				
Think Before You Classify	LLM-based Classificatio n	Financial complaint data	Studied automated classification of financial complaints	Focuses mainly on classification, not the complete complaint-management process

Research gap Identification

Existing Research	Proposed System
Complaint classification only	Complete complaint lifecycle management
No automatic ticket generation	Automatic ticket ID generation
No department mapping	Business Rule Engine for routing
No employee assignment	Intelligent employee allocation
No complaint tracking	Customer complaint tracking portal
No analytics dashboard	Admin dashboard with complaint analytics

Existing studies do not integrate BERT-based complaint classification, business rule automation, and web-based complaint management into a single intelligent banking solution.

Objectives

- To improve the contextual understanding and accuracy of customer complaint classification.
- To reduce delays and inconsistencies caused by manual complaint categorization and routing.
- To enable automated prioritization and department-wise complaint routing.
- To reduce manual effort in complaint handling and improve resolution efficiency.
- To provide a centralized platform for transparent complaint tracking and management.

Innovation

- Context-aware complaint classification using fine-tuned BERT instead of relying only on keyword-based classification.
- Automated routing based on predicted complaint categories through a Business Rule Engine.
- Integration of AI classification with ticket generation, priority assignment, department routing, and employee assignment.

Creativity

- Integration of AI, NLP, Flask, and MySQL into a single platform.
- Role-based web application for Customers, Employees, and Administrators.

Novelty

- Uses Complaint Narrative as the primary input for prediction.
- Predicts four complaint categories simultaneously.
- Combines AI-based classification with automated ticket generation and complaint tracking.

Dataset

Attribute	Details
Dataset Name	Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database
Source	Consumer Financial Protection Bureau (CFPB)
Format	CSV
Domain	Banking & Financial Services
Input Feature	Consumer Complaint Narrative
Target Labels	Product, Sub-product, Issue, Sub-issue

AutoSave complaints.csv - Read-only

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Ready Accessibility: Unavailable

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Date received	Product	Sub-product	Issue	Sub-issue	Consumer compli	Company pub	Company	State	ZIP code	Consumer Submitted	Date sent to Company	Time		
2	29-01-2025	Credit reporting	Credit reporting	Incorrect informat	Account status incorrect		Experian Informat	NY		12543	Web	29-01-2025	In progres	Yes	
3	30-01-2025	Credit reporting	Credit reporting	Improper use of y	Reporting company used your report improperly		Experian Informat	NV		89148	Web	30-01-2025	In progres	Yes	
4	29-01-2025	Payday loan, tit	Installment loan	Problem when making payments			TD BANK US HOLD	FL		33055	N/A	Phone	04-02-2025	In progres	Yes
5	27-01-2025	Credit reporting	Credit reporting	Problem with a co	Their investigation did not fix an error on your		Resurgent Capital	NC		27455	N/A	Postal mail	27-01-2025	In progres	Yes
6	28-01-2025	Credit reporting	Credit reporting	Improper use of y	Reporting company used your report improperly		Experian Informat	IL		60609	Web	28-01-2025	In progres	Yes	
7	28-01-2025	Credit reporting	Credit reporting	Incorrect informat	Information belongs to someone else		Experian Informat	TX		76140	Web	28-01-2025	In progres	Yes	
8	28-01-2025	Credit reporting	Credit reporting	Improper use of y	Reporting company used your report improperly		Experian Informat	GA		30016	Servicem Web	28-01-2025	In progres	Yes	
9	28-01-2025	Mortgage	Conventional hon	Applying for a mor	Trying to communicate with the company to fix		NEXA Mortgage, LI	CA		95370	N/A	Phone	06-02-2025	In progres	Yes
10	13-01-2025	Credit reporting	Credit reporting	Incorrect informat	Information belongs to someone else		Experian Informat	MI		48218	N/A	Phone	06-02-2025	In progres	Yes
11	12-01-2025	Credit reporting	Credit reporting	Incorrect informat	Account inform Subject		Company has	TRANSUNION INTENY		10467	Consent p	Web	12-01-2025	Closed wit	Yes
12	09-01-2025	Debt collection	Credit card debt	Attempts to collec	Debt was paid		Servicing Solutions	IL		60025	Older Am	Web	05-02-2025	Closed wit	Yes
13	15-01-2025	Money transfer	Domestic (US) mo	Other transaction problem			Early Warning Serv	CA		93312	Web	15-01-2025	Closed wit	Yes	
14	17-01-2025	Credit card	General-purpose	Getting a credit ca	Card opened without my consent or knowledge		DISCOVER BANK	NC		28226	N/A	Phone	07-02-2025	Closed wit	Yes
15	11-01-2025	Credit reporting	Credit reporting	Problem with a co	Investigation took more than 30 days		TRANSUNION INTERL			33134	Consent p	Web	11-01-2025	Closed wit	Yes
16	12-01-2025	Credit reporting	Credit reporting	Improper use of y	Reporting com Subject: Follow		Company has	TRANSUNION INTEGA		30102	Consent p	Web	12-01-2025	Closed wit	Yes
17	18-01-2025	Student loan	Federal student lc	Dealing with your	Received bad information about your loan		Servicer under con	MN		55811	Servicem Web	18-01-2025	Untimely / No		
18	20-02-2024	Credit reporting	Credit reporting	Incorrect informat	Information belongs to someone else		EQUIFAX, INC.	CA		90301	Consent n	Web	20-02-2024	Closed wit	Yes
19	26-05-2018	Credit reporting	Credit reporting	Problem with a crs	Their investigation did not fix an error on your		EQUIFAX, INC.	LA		70002	Consent n	Web	26-05-2018	Closed wit	Yes
20	05-04-2012	Student loan	Non-federal stud	Reavine your loan			U.S. BANCORP	MN		55368	N/A	Phone	07-05-2012	Closed wit	Yes

AutoSave complaints.csv - Read-only

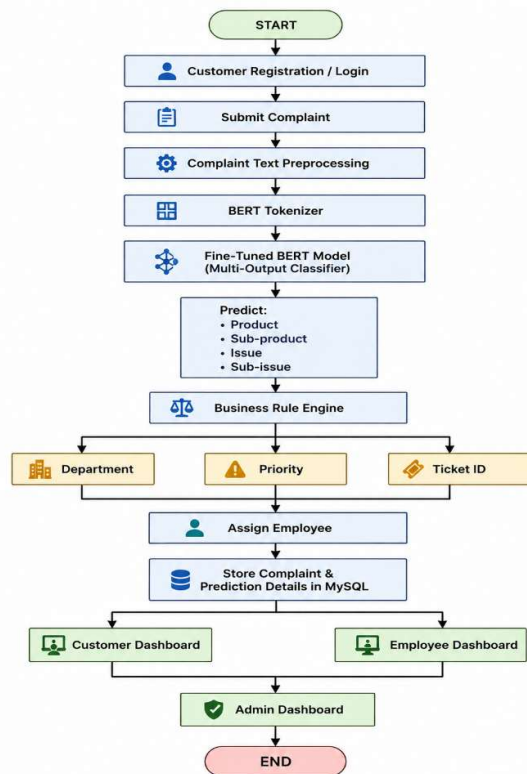
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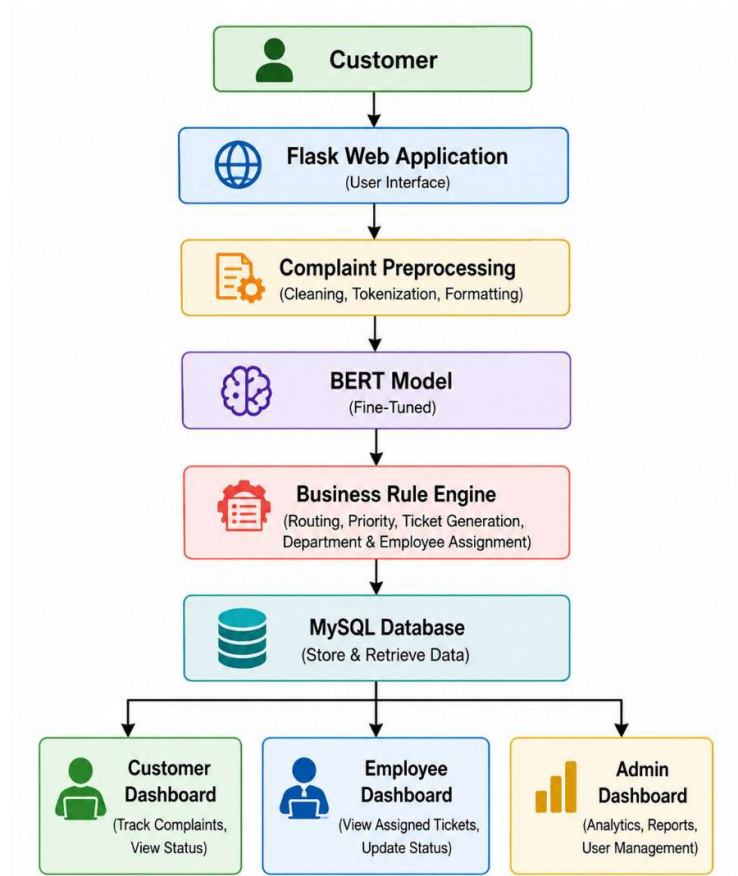
Ready Accessibility: Unavailable

	A	B	C	D	E	F
1	Product	Sub-product	Issue	Sub-issue		
2	Credit reporting or other personal c	Credit reporting	Incorrect information on your report	Account status incorrect		
3	Credit reporting or other personal c	Credit reporting	Improper use of your report	Reporting company used your report improperly		
4	Payday loan, title loan, personal loa	Installment loan	Problem when making payments			
5	Credit reporting or other personal c	Credit reporting	Problem with a company's investigation into an existing	Their investigation did not fix an error on your report		
6	Credit reporting or other personal c	Credit reporting	Improper use of your report	Reporting company used your report improperly		
7	Credit reporting or other personal c	Credit reporting	Incorrect information on your report	Information belongs to someone else		
8	Credit reporting or other personal c	Credit reporting	Improper use of your report	Reporting company used your report improperly		
9	Mortgage	Conventional home mortgage	Applying for a mortgage or refinancing an existing mor	Trying to communicate with the company to fix an issue with the application p		
10	Credit reporting or other personal c	Credit reporting	Incorrect information on your report	Information belongs to someone else		
11	Credit reporting or other personal c	Credit reporting	Incorrect information on your report	Account information incorrect		
12	Debt collection	Credit card debt	Attempts to collect debt not owed	Debt was paid		
13	Money transfer, virtual currency, or Domestic (US) money transfer		Other transaction problem			
14	Credit card	General-purpose credit card or c	Getting a credit card	Card opened without my consent or knowledge		
15	Credit reporting or other personal c	Credit reporting	Problem with a company's investigation into an existing	Investigation took more than 30 days		
16	Credit reporting or other personal c	Credit reporting	Improper use of your report	Reporting company used your report improperly		
17	Student loan	Federal student loan servicing	Dealing with your lender or servicer	Received bad information about your loan		
18	Credit reporting or other personal c	Credit reporting	Incorrect information on your report	Information belongs to someone else		

Proposed Methodology



System Architecture



Feasibility

- Context-aware complaint classification using BERT.
- Reduces manual complaint processing.
- Improves complaint routing accuracy.
- Scalable and easy to integrate with banking systems.
- User-friendly dashboards for all stakeholders.

Feasibility Analysis

Type	Description
Technical	Uses proven AI and web technologies
Economic	Open-source tools minimize development cost
Operational	Easy to use and maintain
Schedule	Can be completed within the project timeline

Project Plan

Phase	Activity
Phase 1	Literature Survey
Phase 2	Dataset Collection & Preprocessing
Phase 3	BERT Model Development
Phase 4	Flask Web Application Development
Phase 5	Database Integration
Phase 6	Business Rule Engine Development
Phase 7	Testing & Evaluation
Phase 8	Documentation & Final Deployment

References

- CFPB Consumer Complaint Database – <https://www.consumerfinance.gov/data-research/consumer-complaints/>
- Devlin et al., BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding – <https://arxiv.org/abs/1810.04805>
- Bastani et al., LDA for Topic Modeling of CFPB Consumer Complaints – <https://arxiv.org/abs/1807.07468>
- Sangaraju et al., Topic Modelling on CFPB Data Using BERT Based Embeddings – <https://arxiv.org/abs/2205.07259>
- Vaishnav et al., Predictive Analysis of CFPB Consumer Complaints Using Machine Learning – <https://arxiv.org/abs/2407.06399>